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IMMERSION COOLING SYSTEMS AND METHODS FOR TRACTION BATTERY PACK SYSTEMS

Abstract

Immersion cooling systems are provided for managing thermal energy levels within a traction battery pack system. An exemplary immersion cooling system may include a flow control valve that is configured to control a flow of a cooling fluid (e.g., a dielectric fluid) through either a primary closed loop cooling circuit or a secondary closed loop cooling circuit of the immersion cooling system for thermally managing a battery module of a battery pack assembly. A control module may control a position of the flow control valve based at least on a temperature of the cooling fluid exiting the battery pack assembly. When the flow control valve directs the cooling fluid through the secondary closed loop cooling circuit, a portion of the primary closed loop cooling circuit is reserved for providing a dedicated gas exit flow path for expelling battery vent byproducts from the battery pack assembly.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to electrified vehicle traction battery pack systems, and more particularly to immersion cooling systems capable of managing battery cell thermal energy levels across multiple operating conditions of the traction battery pack system.

BACKGROUND

[0002] An electrified vehicle includes a traction battery pack for powering electric machines and other electrical loads of the vehicle. The traction battery pack includes a plurality of battery cells and various other battery internal components that support electric vehicle propulsion.

SUMMARY

[0003] A traction battery pack system according to an exemplary aspect of the present disclosure includes, among other things, a battery pack assembly including a battery module housed within an enclosure assembly, a primary closed loop cooling circuit that establishes a first flow path of an immersion cooling system, a secondary closed loop cooling circuit that establishes a second flow path of the immersion cooling system, a flow control valve arranged to control a flow of a cooling fluid along either the first flow path or the second flow path, and a control module programmed to control a position of the flow control valve based on a temperature of the cooling fluid exiting the battery pack assembly.

[0004] In a further non-limiting embodiment of the foregoing traction battery pack system, a temperature sensor is positioned within or near an outlet port of the battery pack assembly.

[0005] In a further non-limiting embodiment of either of the foregoing traction battery pack systems, the cooling fluid is a dielectric fluid.

[0006] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the control module is programmed to command the flow control valve to a first position for directing the cooling fluid along the first flow path when the temperature of the cooling fluid is less than or equal to a predefined temperature threshold, and is further programmed to command the flow control valve to a second position for directing the cooling fluid along the second flow path when the temperature of the cooling fluid is greater than the predefined temperature threshold.

[0007] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the first flow path extends through both a first battery internal fluid passageway and a second battery internal fluid passageway of the battery pack assembly. The second flow path extends through the first battery internal fluid passageway but not the second battery internal fluid passageway of the battery pack assembly.

[0008] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the second battery internal fluid passageway establishes a dedicated gas exit flow path for expelling a battery vent byproduct from the battery pack assembly when the temperature of the cooling fluid is greater than the predefined temperature threshold.

[0009] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the first battery internal fluid passageway extends between bottom sides of a plurality of battery cells of the battery module and an enclosure tray of the enclosure assembly, and the second battery internal fluid passageway extends between top sides of the plurality of battery cells of the battery module and an enclosure cover of the enclosure assembly.

[0010] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the primary closed loop cooling circuit of the immersion cooling system includes a first inlet port, a first battery internal fluid passageway, a first outlet port, a second inlet port, an external fluid passageway that connects between the first outlet port and the second inlet port, a second battery internal fluid passageway, and a second outlet port.

[0011] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the secondary closed loop cooling circuit of the immersion cooling system includes the first inlet port, the first battery internal fluid passageway, the first outlet port, and a return line.
[0012] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the immersion cooling system further includes a reservoir that is fluidly connected to the first inlet port.
[0013] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the immersion cooling system further includes a pump arranged between the reservoir and the first inlet port.
[0014] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the reservoir is positioned on an opposite side of the battery pack assembly from the flow control valve.
[0015] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, a first heat exchanger is fluidly connected to the external fluid passageway, and a second heat exchanger is fluidly connected to the return line.
[0016] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, a first position of the flow control valve is configured to divert the cooling fluid into the external fluid passageway of the primary closed loop cooling circuit, and a second position of the flow control valve is configured to divert the cooling fluid into the return line of the secondary closed loop cooling circuit.
[0017] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the immersion cooling system further includes a liquid-gas separator fluidly connected to the second outlet port.
[0018] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the liquid-gas separator is positioned on an opposite side of the battery pack assembly from the flow control valve.
[0019] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, a temperature sensor is positioned within or near the second outlet port.
[0020] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the flow control valve is positioned outside of the enclosure assembly of the battery pack assembly.
[0021] In a further non-limiting embodiment of any of the foregoing traction battery pack systems, the flow control valve is positioned between an outlet port of the battery pack assembly and a heat exchanger of the primary closed loop cooling circuit.
[0022] A method according to another exemplary aspect of the present disclosure includes, among other things, circulating a cooling fluid along a first flow path provided by a primary closed loop cooling circuit of an immersion cooling system, monitoring a temperature of the cooling fluid as it travels along the first flow path, and when the temperature exceeds a predefined temperature threshold, circulating the cooling fluid along a second flow path provided by a secondary closed loop cooling circuit of the immersion cooling system. Circulating the cooling fluid along the second flow path includes preventing the cooling fluid from passing through a portion of the first flow path of the primary closed loop cooling circuit.
[0023] The embodiments, examples, and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.
[0024] The various features and advantages of this disclosure will become apparent to those skilled in the art from the following detailed description. The drawings that accompany the detailed description can be briefly described as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. **1** schematically illustrates an electrified vehicle.

[0026] FIG. **2** schematically illustrates an immersion cooling system of a traction battery pack system.

[0027] FIG. **3** schematically illustrates a first operation mode of the immersion cooling system of FIG. **2**.

[0028] FIG. **4** schematically illustrates a second operation mode of the immersion cooling system of FIG. **2**.

DETAILED DESCRIPTION

[0029] This disclosure details immersion cooling systems for managing thermal energy levels within a traction battery pack system. An exemplary immersion cooling system may include a flow control valve that is configured to control a flow of a cooling fluid (e.g., a dielectric fluid) through either a primary closed loop cooling circuit or a secondary closed loop cooling circuit of the immersion cooling system for thermally managing a battery module of a battery pack assembly. A control module may control a position of the flow control valve based at least on a temperature of the cooling fluid exiting the battery pack assembly. When the flow control valve directs the cooling fluid through the secondary closed loop cooling circuit, a portion of the primary closed loop cooling circuit is reserved for providing a dedicated gas exit flow path for expelling battery vent byproducts from the battery pack assembly. These and other features are discussed in greater detail in the following paragraphs of this detailed description.

[0030] FIG. **1** schematically illustrates an electrified vehicle **10**. The electrified vehicle **10** may include any type of electrified powertrain. In an embodiment, the electrified vehicle **10** is a battery electric vehicle (BEV). However, the concepts described herein are not limited to BEVs and could extend to other electrified vehicles, including, but not limited to, hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEV's), fuel cell vehicles, etc. Therefore, although not specifically shown in the exemplary embodiment, the powertrain of the electrified vehicle **10** could be equipped with an internal combustion engine that can be employed either alone or in combination with other power sources to propel the electrified vehicle **10**.

[0031] In the illustrated embodiment, the electrified vehicle **10** is depicted as a car. However, the electrified vehicle **10** could alternatively be a sport utility vehicle (SUV), a van, a pickup truck, or any other vehicle configuration. Although a specific component relationship is illustrated in the figures of this disclosure, the illustrations are not intended to limit this disclosure. The placement and orientation of the various components of the electrified vehicle **10** are shown schematically and could vary within the scope of this disclosure. In addition, the various figures accompanying this disclosure are not necessarily drawn to scale, and some features may be exaggerated or minimized to emphasize certain details of a particular component or system.

[0032] In the illustrated embodiment, the electrified vehicle **10** is a full electric vehicle propelled solely through electric power, such as by one or more electric machines **12**, without assistance from an internal combustion engine. The electric machine **12** may operate as an electric motor, an electric generator, or both. The electric machine **12** receives electrical power and can convert the electrical power to torque for driving one or more wheels **14** of the electrified vehicle **10**.

[0033] A voltage bus **16** may electrically couple the electric machine **12** to a traction battery pack system **18**. The traction battery pack system **18** is an exemplary electrified vehicle battery. The traction battery pack system **18** may include a high voltage traction battery pack assembly that includes a plurality of battery cells capable of outputting electrical power to power the electric machine **12** and/or other electrical loads of the electrified vehicle **10**. Other types of energy storage devices and/or output devices could alternatively or additionally be used to electrically power the electrified vehicle **10**.

[0034] The traction battery pack system **18** may be secured to an underbody **20** of the electrified vehicle **10**. However, the traction battery pack system **18** could be located elsewhere on the electrified vehicle **10** within the scope of this disclosure.

[0035] FIG. **2** illustrates additional details associated with the traction battery pack system **18** of the electrified vehicle **10** of FIG. **1**. The traction battery pack system **18** may include one or more battery modules **22** (e.g., battery assemblies or groupings of rechargeable battery cells **24**) capable of outputting electrical power to power the electric machine **12** and/or other electrical loads of the electrified vehicle **10**.

[0036] The battery cells **24** may be stacked side-by-side along a stack axis to construct a grouping of battery cells **24**, sometimes referred to as a “cell stack.” The total number of battery modules **22** and battery cells **24** provided within the traction battery pack system **18** is not intended to limit this disclosure.

[0037] In an embodiment, the battery cells **24** of each battery module **22** are prismatic, lithium-ion cells. However, battery cells having other geometries (cylindrical, pouch, etc.), other chemistries (nickel-metal hydride, lead-acid, etc.), or both could alternatively be utilized within the scope of this disclosure.

[0038] The battery modules **22** and various other battery internal components (e.g., bussed electrical center, battery electric control module, wiring, connectors, etc.) may be housed inside of an enclosure assembly **28** of the traction battery pack system **18**. Together, the battery modules **22** and the enclosure assembly **28** may establish a battery pack assembly **25** of the traction battery pack system **18**.

[0039] The battery modules **22** may be arranged in one or more rows inside the enclosure assembly **28**. Although only a single battery module **22** having twelve battery cells **24** is shown in FIG. **2**, other configurations are possible. Accordingly, it should be appreciated that the traction battery pack system **18** could include a greater number of battery modules **22** and/or a greater or fewer number of battery cells **24** within the scope of this disclosure.

[0040] Although shown schematically, the enclosure assembly **28** may embody a multi-piece design that includes an enclosure cover **26** and an enclosure tray **30** that are joined together to establish an interior for housing the battery module(s) **22**. The size, shape, and overall configuration of the enclosure assembly **28** is not intended to limit this disclosure. In an embodiment, the enclosure assembly **28** provides a sealed enclosure around the battery module(s) **22** and other battery internal components of the battery pack assembly **25**.

[0041] The traction battery pack system **18** may additionally include an immersion cooling system **32**. As further detailed below, the immersion cooling system **32** may include a primary closed loop cooling circuit **34** and a secondary closed loop cooling circuit **36** arranged for thermally managing the battery module **22** and for managing battery vent gases and other byproducts during different operating conditions of the traction battery pack system **18**. Although schematically shown, the various subcomponents of the immersion cooling system **32** can be fluidly interconnected by various conduits or passages such as tubes, hoses, pipes, etc.

[0042] The immersion cooling system **32** may be configured for directly contacting individual surfaces of the battery cells **24** by circulating a cooling fluid F along a flow path established by either the primary closed loop cooling circuit **34** or the secondary closed loop cooling circuit **36**. In an embodiment, the cooling fluid F is a dielectric fluid. However, other fluids could be utilized within the scope of this disclosure.

[0043] A flow control valve **38** may be configured to control the flow of the cooling fluid F along the respective flow path provided by either the primary closed loop cooling circuit **34** or the secondary closed loop cooling circuit **36** depending on a current operating condition of the traction battery pack system **18**. The flow control valve **38** may be a multi-position solenoid valve (e.g., a three-way valve) or any other suitable type of electronically controllable valve.

[0044] The primary closed loop cooling circuit **34** of the immersion cooling system **32** may include a first inlet port **40**, a first battery internal fluid passageway **42**, a first outlet port **44**, a second inlet port **46**, a second battery internal fluid passageway **48**, and a second outlet port **50**. The first battery internal fluid passageway **42** may extend between bottom sides **54** of the battery cells **24** of the battery module **22** and the enclosure tray **30**, and the second battery internal fluid passageway **48** may extend between top sides **52** of the battery cells **24** of the battery module **22** and the enclosure cover **26**. An external fluid passageway **56** may connect between the first outlet port **44** and the second inlet port **46**. The external fluid passageway **56** may be fluidly connected to the flow control valve **38** and to a first heat exchanger **58** (e.g., a first fluid-to-air heat exchanger), which are both disposed outside of the enclosure assembly **28** of the battery pack assembly **25**.
[0045] The first inlet port **40** may be fluidly connected to a reservoir **60** that is configured for storing the cooling fluid F. A pump **62** may be operated to selectively circulate the cooling fluid F through the respective flow path of either the primary closed loop cooling circuit **34** or the secondary closed loop cooling circuit **36**. In an embodiment, the pump **62** is located between the reservoir **60** and the first inlet port **40**. However, the pump **62** could be located elsewhere within the scope of this disclosure. The pump **62** may be an electrically powered fluid pump or another type of pump within the scope of this disclosure.
[0046] The reservoir **60** and the pump **62** may both be located outside of the enclosure assembly **28** of the battery pack assembly **25**. In an embodiment, the reservoir **60** and the pump **62** are located on an opposite side of the enclosure assembly **28** from the flow control valve **38** and the first heat exchanger **58**.
[0047] The secondary closed loop cooling circuit **36** of the immersion cooling system **34** may include the first inlet port **40**, the first battery internal fluid passageway **42**, the first outlet port **44**, and a return line **72**. Some pathways (e.g., the first inlet port **40**, the first battery internal fluid passageway **42**, and the first outlet port **44**) of the immersion cooling system **34** are thus shared as part of both the primary closed loop cooling circuit **34** and the secondary closed loop cooling circuit **36**. The return line **72** may connect between the flow control valve **38** and the reservoir **60**. A second heat exchanger **74** (e.g., a second fluid-to-air heat exchanger) may be provided within the return line **72**. The return line **72** and the second heat exchanger **74** are both disposed outside of the enclosure assembly **28** of the battery pack assembly **25**.
[0048] When the cooling fluid F is circulated along a flow path provided by the primary closed loop cooling circuit **34**, the pump **62** may be operated to selectively force the cooling fluid F through the first inlet port **40** and then through the first battery internal fluid passageway **42**. The cooling fluid F may pick up heat from the battery cells **24** through convective heat transfer as it flows across the first battery internal fluid passageway **42**. The cooling fluid F may then exit the battery pack assembly **25** through the first outlet port **44**. The flow control valve **38** may then direct the cooling fluid F into the external fluid passageway **56**. The cooling fluid F may then pass through the first heat exchanger **58** from within the external fluid passageway **56**. Thermal energy picked up from the battery cells **24** while passing through the first battery internal fluid passageway **42** may be transferred from the cooling fluid F to ambient air within the first heat exchanger **58**.
[0049] The cooling fluid F may then flow through the second inlet port **46** prior to entering the second battery internal fluid passageway **48**. The cooling fluid F may pick up additional heat from the battery cells **24** through convective heat transfer as it flows through the second battery internal fluid passageway **48**. The cooling fluid F may then exit the second battery internal fluid passageway **48** through the second outlet port **50**.
[0050] The second outlet port **50** may be fluidly connected to a liquid-gas separator **64**. Gas may be separated from the cooling fluid F within the liquid-gas separator **64** and may be exhausted to atmosphere through a gas outlet **66** as schematically shown at arrow **68**. As schematically shown, the liquid-gas separator **64** may be located outside of the enclosure assembly **28** of the battery pack assembly **25** and may be positioned on an opposite side of the enclosure assembly **28** from the first

heat exchanger **58** and the flow control valve **38**. The degassed cooling fluid F may be returned to the reservoir **60** within a connection line **70** that is fluidly connected to both the liquid-gas separator **64** and the reservoir **60**.

[0051] When the cooling fluid F is circulated along a flow path provided by the secondary closed loop cooling circuit **36**, the pump **62** may be operated to selectively force the cooling fluid F through the first inlet port **40** and then into the first battery internal fluid passageway **42**. The cooling fluid F may pick up heat from the battery cells **24** through convective heat transfer as it flows through the first battery internal fluid passageway **42**. The cooling fluid F may exit the first battery internal fluid passageway **42** through the first outlet port **44**. The flow control valve **38** may then direct the cooling fluid F into the return line **72**. The cooling fluid F may pass through the second heat exchanger **74** from within the return line **72**. Thermal energy picked up from the battery cells **24** while passing through the first battery internal fluid passageway **42** may be transferred from the cooling fluid F to ambient air within the second heat exchanger **74** prior to returning the cooling fluid F to the reservoir **60**. Notably, the cooling fluid F does not pass through the second battery internal fluid passageway **48** while being circulated through the secondary closed loop cooling circuit **36**.

[0052] The immersion cooling system **32** may further include one or more temperature sensors **76** and a control module **78**. The temperature sensor **76** may be configured to sense the temperature of the cooling fluid F. In an embodiment, the temperature sensor **76** is provided within or near the second outlet port **50**. The temperature sensor **76** may therefore be arranged to sense the temperature of the cooling fluid F exiting the enclosure assembly **28** of the battery pack assembly **25**.

[0053] The control module **78** may be operably connected to the pump **62**, the flow control valve **38**, and the temperature sensor **76** and may be programmed to control operations of the immersion cooling system **32** for thermally managing the battery module **22** of the traction battery pack system **18**. The control module **78** may include both hardware and software and could be part of an overall vehicle control system, such as a vehicle system controller (VSC), or could alternatively be a stand-alone controller or collection of controllers that are separate from the VSC. It should therefore be understood that the control module **78** and one or more additional controllers operably coupled thereto can collectively be referred to as a “control module” within the scope of this disclosure.

[0054] The control module **78** may be programmed with executable instructions for interfacing with and commanding operation of various components of the immersion cooling system **32** as part of a control strategy for controlling the flow path of the cooling fluid F. The control module **78** may include a processor **80** and non-transitory memory **82** for executing the various control strategies and modes associated with the immersion cooling system **32**. The processor **80** may be a custom made or commercially available processor, a central processing unit (CPU), or generally any device for executing software instructions. The memory **82** may include any one or combination of volatile memory elements and/or nonvolatile memory elements. The processor **80** may be operably coupled to the memory **82** and may be configured to execute one or more programs stored in the memory **82** based on the various inputs received from other devices (e.g., inputs from the temperature sensor **76**) associated with the immersion cooling system **32**.

[0055] The temperature sensor **76** may be configured to periodically provide input signals to the control module **78** that are indicative of the temperature of the cooling fluid F exiting the battery pack assembly **25** through the second outlet port **50**. In response to receiving the input signals, the control module **78** may control a position of the flow control valve **38** in order to direct the cooling fluid F along a desired flow path for thermally managing the battery cells **24** and/or for responding to a battery thermal event that could occur within one or more of the battery cells **24**. A battery thermal event may occur, for example, during over-charging conditions, over-discharging conditions, or other conditions and can cause one or more of the battery cells **24** to expel battery vent byproducts, which can include gases, effluent particles, and/or other vent byproducts.

[0056] A first operation mode of the immersion cooling system **32** that can be commanded by the control module **78** is schematically illustrated in FIG. **3**. The first operation mode may be considered a default mode of the immersion cooling system **32** and can occur when the temperature input signals received from the temperature sensor **76** indicate a temperature of the cooling fluid F that is less than or equal to a predefined temperature threshold (e.g., about 150 degrees C. or some other suitable temperature selected based on design specific factors). When this occurs, the control module **78** may command the flow control valve **38** to a first position for directing the cooling fluid F along a first flow path **99** provided by the primary closed loop cooling circuit **34**. The cooling fluid F may therefore flow through both the first battery internal fluid passageway **42** and the second battery internal fluid passageway **48** for thermally managing the battery cells **24** of the battery module **22** during normal operating conditions of the traction battery pack system **18**. Notably, flow of the cooling fluid F is blocked from entering the return line **72** of the secondary closed loop cooling circuit **36** during the first operation mode.

[0057] A second operation mode of the immersion cooling system **32** that can be commanded by the control module **78** is schematically illustrated in FIG. **4**. The second operation mode may be specifically adapted for responding to a battery thermal event that could occur within one or more of the battery cells **24** of the battery module **22**. For example, when the temperature input signals received from the temperature sensor **76** indicate a temperature that is greater than or equal to the predefined temperature threshold (e.g., about 150 degrees C. or some other suitable temperature selected based on design specific factors), the control module **78** may command the flow control valve **38** to a second position for directing the cooling fluid F along a second flow path **199** provided by the secondary closed loop cooling circuit **36**. The cooling fluid F may therefore flow through only the first battery internal fluid passageway **42** and not the second battery internal fluid passageway **48** for thermally managing the battery cells **24**. Notably, flow of the cooling fluid F is blocked from entering the external fluid passageway **56** of the primary closed loop cooling circuit **34** by the flow control valve **38** during the second operation mode.

[0058] Since the cooling fluid F is not circulated through the second battery internal fluid passageway **48** during the second operation mode, the second battery internal fluid passageway **48** may provide a dedicated gas exit flow path **299** for expelling battery vent byproducts V released by one or more of the battery cells **24** from the battery pack assembly **25** during a battery thermal event. An excessive pressure build-up inside the enclosure assembly **28** of the battery pack assembly **25** may therefore be prevented during the battery thermal event that necessitated operation of the immersion cooling system in the second operation mode.

[0059] The battery vent byproducts V may travel through the second outlet port **50** from the second internal fluid passageway **48**. The battery vent byproducts V may be exhausted to atmosphere through the gas outlet **66** of the liquid-gas separator **64** as schematically shown at arrow **84**.

[0060] Once the battery vent byproducts V have been expelled from the battery pack assembly **25** and the temperature sensed by the temperature sensor **76** drops to being less than or equal to the predefined temperature threshold, the control module **78** may command the flow control valve **38** back to the first position for directing the cooling fluid F through the primary closed loop cooling circuit **34**. Normal operation of the immersion cooling system **32** may therefore be re-established following the battery thermal event.

[0061] The control module **78** may be programmed to control the immersion cooling system **32** in the second operation mode for a predefined amount of time (e.g., about **20** seconds or some other suitable amount of time selected based on design specific factors). The predefined amount of time is a sufficient amount of time to allow the battery vent byproducts V to be expelled from the battery pack assembly **25** without creating an excessive pressure situation inside the enclosure assembly **28**.

[0062] The exemplary traction battery packs of this disclosure include an immersion cooling

system for providing enhanced battery cell thermal management and vent gas management. A flow path of a cooling fluid used within the immersion cooling system may be controlled based on a temperature of the cooling fluid, thereby reserving battery internal space for providing a gas exit flow path for expelling vent gases from the battery pack and preserving thermal performance during battery thermal events.
[0063] Although the different non-limiting embodiments are illustrated as having specific components or steps, the embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from any of the non-limiting embodiments in combination with features or components from any of the other non-limiting embodiments.
[0064] It should be understood that like reference numerals identify corresponding or similar elements throughout the several drawings. It should be understood that although a particular component arrangement is disclosed and illustrated in these exemplary embodiments, other arrangements could also benefit from the teachings of this disclosure.
[0065] The foregoing description shall be interpreted as illustrative and not in any limiting sense. A worker of ordinary skill in the art would understand that certain modifications could come within the scope of this disclosure. For these reasons, the following claims should be studied to determine the true scope and content of this disclosure.

Claims

1. A traction battery pack system, comprising: a battery pack assembly including a battery module housed within an enclosure assembly; a primary closed loop cooling circuit that establishes a first flow path of an immersion cooling system; a secondary closed loop cooling circuit that establishes a second flow path of the immersion cooling system; a flow control valve arranged to control a flow of a cooling fluid along either the first flow path or the second flow path; and a control module programmed to control a position of the flow control valve based on a temperature of the cooling fluid exiting the battery pack assembly.
2. The traction battery pack system as recited in claim 1, comprising a temperature sensor positioned within or near an outlet port of the battery pack assembly.
3. The traction battery pack system as recited in claim 1, wherein the cooling fluid is a dielectric fluid.
4. The traction battery pack system as recited in claim 1, wherein the control module is programmed to command the flow control valve to a first position for directing the cooling fluid along the first flow path when the temperature of the cooling fluid is less than or equal to a predefined temperature threshold and is further programmed to command the flow control valve to a second position for directing the cooling fluid along the second flow path when the temperature of the cooling fluid is greater than the predefined temperature threshold.
5. The traction battery pack system as recited in claim 4, wherein the first flow path extends through both a first battery internal fluid passageway and a second battery internal fluid passageway of the battery pack assembly, and further wherein the second flow path extends through the first battery internal fluid passageway but not the second battery internal fluid passageway of the battery pack assembly.
6. The traction battery pack system as recited in claim 5, wherein the second battery internal fluid passageway establishes a dedicated gas exit flow path for expelling a battery vent byproduct from the battery pack assembly when the temperature of the cooling fluid is greater than the predefined temperature threshold.
7. The traction battery pack system as recited in claim 5, wherein the first battery internal fluid passageway extends between bottom sides of a plurality of battery cells of the battery module and an enclosure tray of the enclosure assembly, and the second battery internal fluid passageway

extends between top sides of the plurality of battery cells of the battery module and an enclosure cover of the enclosure assembly.
8. The traction battery pack system as recited in claim 1, wherein the primary closed loop cooling circuit of the immersion cooling system includes a first inlet port, a first battery internal fluid passageway, a first outlet port, a second inlet port, an external fluid passageway that connects between the first outlet port and the second inlet port, a second battery internal fluid passageway, and a second outlet port.
9. The traction battery pack system as recited in claim 8, wherein the secondary closed loop cooling circuit of the immersion cooling system includes the first inlet port, the first battery internal fluid passageway, the first outlet port, and a return line.
10. The traction battery pack system as recited in claim 9, wherein the immersion cooling system further includes a reservoir that is fluidly connected to the first inlet port.
11. The traction battery pack system as recited in claim 10, wherein the immersion cooling system further includes a pump arranged between the reservoir and the first inlet port.
12. The traction battery pack system as recited in claim 10, wherein the reservoir is positioned on an opposite side of the battery pack assembly from the flow control valve.
13. The traction battery pack system as recited in claim 9, comprising a first heat exchanger fluidly connected to the external fluid passageway, and a second heat exchanger fluidly connected to the return line.
14. The traction battery pack system as recited in claim 9, wherein a first position of the flow control valve is configured to divert the cooling fluid into the external fluid passageway of the primary closed loop cooling circuit, and a second position of the flow control valve is configured to divert the cooling fluid into the return line of the secondary closed loop cooling circuit.
15. The traction battery pack system as recited in claim 8, wherein the immersion cooling system further includes a liquid-gas separator fluidly connected to the second outlet port.
16. The traction battery pack system as recited in claim 15, wherein the liquid-gas separator is positioned on an opposite side of the battery pack assembly from the flow control valve.
17. The traction battery pack system as recited in claim 8, comprising a temperature sensor positioned within or near the second outlet port.
18. The traction battery pack system as recited in claim 1, wherein the flow control valve is positioned outside of the enclosure assembly of the battery pack assembly.
19. The traction battery pack system as recited in claim 18, wherein the flow control valve is positioned between an outlet port of the battery pack assembly and a heat exchanger of the primary closed loop cooling circuit.
20. A method, comprising: circulating a cooling fluid along a first flow path provided by a primary closed loop cooling circuit of an immersion cooling system; monitoring a temperature of the cooling fluid as it travels along the first flow path; and when the temperature exceeds a predefined temperature threshold, circulating the cooling fluid along a second flow path provided by a secondary closed loop cooling circuit of the immersion cooling system, wherein circulating the cooling fluid along the second flow path includes preventing the cooling fluid from passing through a portion of the first flow path of the primary closed loop cooling circuit.